

同济大学电子与信息工程学院

课 程 介 绍

课程编号	100433	学分	2
课程名称(中)	计算机图形学		
课程名称(英)	Computer Graphics		
总学时	34	周学时	2
实验学时		上机学时	17(课外)
先修课程 (参照教学大纲所列课程)	线性代数、高级语言程序设计、数据结构		
课程中文简介 (200-300字)	《计算机图形学》是计算机科学技术系开设的专业大类平台（C2类）课程。通过本课程的开设和教学实践，使学生了解计算机图形学的发展，掌握图形学的基本原理、算法和实现技术。重点学习各种光栅图形的生成、参数曲线曲面和实体的几何造型，以及几何变换、裁剪、消隐、明暗处理和真实感图形生成等算法，了解纹理映射、阴影生成和反走样等技术，学会基本的图形软件开发技术，以满足各种应用领域对图形处理的要求。		
课程英文简介	“Computer Graphics” (CG) is one of optional courses offered for undergraduate students of professional category C2 classes by the department of computer science and technology. By setting up this course, students are able to know the historical development of CG discipline well, and grasp various 2D and 3D graphic principles, algorithms and implementation techniques. Topics mainly covered in the teaching process include raster scan conversion, 2-dimensional transformations and clipping, parametric curves and surfaces, 3-dimensional display techniques, representations and transformations, hidden-surface elimination, lighting and shading models, with some other essentials of texture mapping, shadow generation and antialiasing techniques. There will be a strong emphasis on the mathematical and geometric aspects of graphics, and programming knowledge and experiences of graphic algorithms for further applications.		